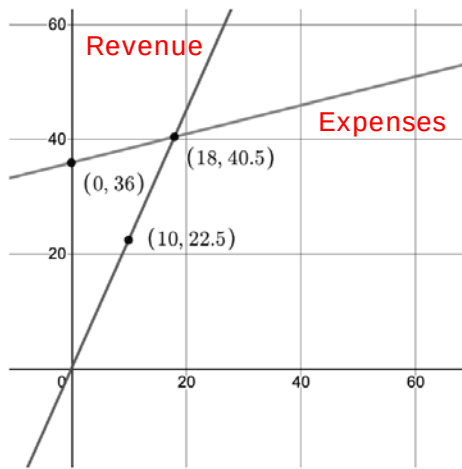


Grade 8
Unit 8
Lesson 10 Practice

Name Answer Key
Date _____
Period _____

Elizabeth and her friends decided to bake and sell cookies after school to earn some extra spending money. After the initial expenses of \$36, they purchase the ingredients for \$0.25 per cookie. They will sell each cookie for \$2.25.

The system $\begin{cases} y = 2.25x \\ y = 0.25x + 36 \end{cases}$ represents this situation, where x is the number of cookies and y is cost. The graph of the system is shown with three points labeled.



1. Label the line that represents the expenses.
2. Label the line that represents the revenue.
3. What is the solution to the system?
(18,40.5)
4. Explain the meaning of the solution in this context.
Once 18 cookies are baked and sold, the expenses will be equal to the revenue: \$40.50.

Andy's Cab Service charges a \$6 fee plus \$0.50 per mile. His twin brother Randy starts a rival business where he charges \$0.80 per mile but does not charge a fee.

The system $\begin{cases} y = \frac{1}{2}x + 6 \\ y = \frac{4}{5}x \end{cases}$ represents this situation, where x is the number of miles and y is the cost.

5. Graph the system.

6. Label the axes with the appropriate titles.

7. Label each line with "Andy" or "Randy".

8. What is the solution to the system? Circle it on the graph.
(20,16)

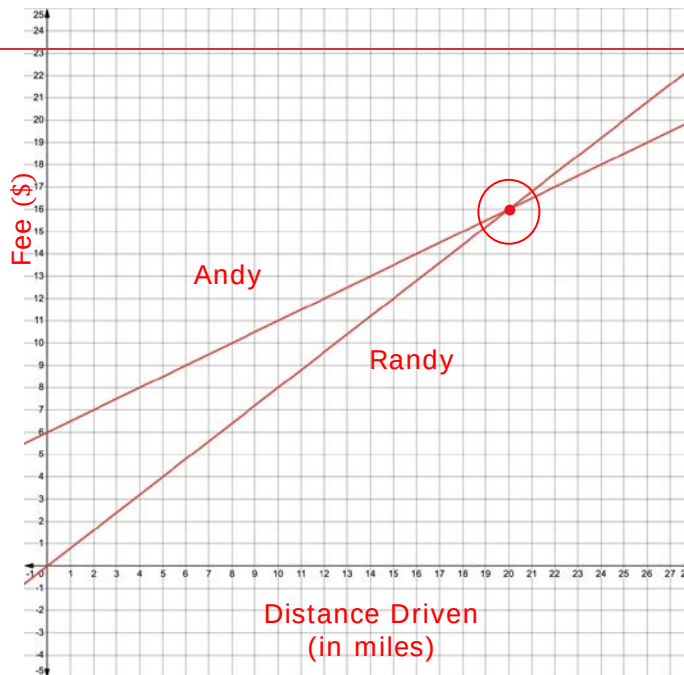
9. Explain the meaning of the solution in the context?

After driving a passenger for 20 miles, both Andy and Randy charge the same fee of \$16.

10. For what trip distances should a customer use Andy's Cab Service? For what trip distances should a customer use Randy's Cab Service?

Answers may vary.

Andy's should be used for distances greater than 20 miles. Randy's should be used for distances less than 20 miles.



Commented [JP1]: <https://www.desmos.com/calculator/n2f3ki4w0w>